

CR-007: Course Seat Capacity and Availability

1. Change Request Details

Change Request ID: CR-007

Title: Course Seat Capacity and Availability

Priority: High

Difficulty: Intermediate

Change Type: Database, backend and frontend change

2. Business Requirement

Courses should support an optional maximum number of students. The system must prevent over-enrollment and clearly display course availability to students and administrators.

3. Current Behaviour

- Courses currently have unlimited capacity.
- The database does not store a maximum student capacity.
- The system does not calculate or display available seats.
- Students can continue enrolling without a course capacity restriction.
- Concurrent enrollment attempts are not protected against exceeding a capacity limit.
- Administrators can currently change course information without a capacity-based enrollment restriction.

4. Requested Change

Add optional seat capacity to courses.

The application must:

- Add a `max_students INT NULL` column to courses.
- Treat `NULL` as unlimited capacity.
- Require `max_students` to be a positive integer when provided.
- Add a Maximum Students field to the Admin Course form.
- Treat a blank Maximum Students field as unlimited capacity.
- Return `max_students` in course responses.
- Return `enrolled_count` in course responses.
- Return `seats_remaining` in course responses.
- Return `is_full` in course responses.
- Display course availability on the Home page.
- Display course availability in the Course Catalogue.

- Display course availability on Course Details.
- Display course availability in Admin Course Management.
- Display capacity in the format `15 / 30 students`.
- Display `Unlimited` when no capacity is configured.
- Prevent enrollment when the course has reached its capacity.
- Return HTTP 409 when a course is full.
- Prevent administrators from reducing capacity below the current approved enrollment count.
- Allow administrators to set capacity equal to the current approved enrollment count.
- Perform the capacity check and enrollment insertion atomically.
- Prevent concurrent enrollment requests from exceeding the configured capacity.
- Free a seat when an enrollment is deleted.
- Preserve unlimited-capacity behaviour for existing courses.
- Seed at least one course with a small capacity for testing.

5. Scope

This change applies to course capacity management, course availability calculation, enrollment validation, transactional enrollment, course administration, course catalogue display, Course Details, Home page availability, and related frontend feedback.

The change includes:

- Adding the `max_students` database column.
- Creating a database migration.
- Updating the database schema.
- Updating seed data with at least one limited-capacity course.
- Validating course capacity values.
- Supporting unlimited capacity through `NULL`.
- Adding Maximum Students to the Admin Course form.
- Calculating approved enrollment count.
- Calculating remaining seats.
- Calculating whether a course is full.
- Returning capacity information through course APIs.
- Displaying availability across required frontend pages.
- Preventing enrollment when capacity has been reached.
- Returning HTTP 409 for full courses.
- Implementing atomic capacity checking and enrollment insertion.
- Protecting against concurrent enrollment attempts.
- Preventing administrators from reducing capacity below approved enrollment.
- Allowing capacity to equal the current approved enrollment count.
- Freeing seats when enrollments are deleted.
- Maintaining existing unlimited-capacity courses.
- Displaying a clear `Course Full` state.
- Distinguishing a full course from a course in which the current student is already enrolled.

The change does not include:

- Removing support for unlimited-capacity courses.
- Changing unrelated course functionality.
- Changing unrelated enrollment functionality.
- Allowing negative, zero, or non-integer course capacities.
- Allowing administrators to reduce capacity below the current approved enrollment count.

- Relying only on frontend checks to prevent over-enrollment.
- Allowing concurrent requests to bypass capacity restrictions.

6. Functional Requirements

6.1 Database and Course Capacity

FR-001: Maximum Students Column

The courses table must include:

```
max_students INT NULL
```

FR-002: Optional Capacity

```
max_students may be NULL.
```

A `NULL` value means the course has unlimited capacity.

FR-003: Positive Capacity

When `max_students` is provided, it must be a positive integer.

Values such as:

- 0
- Negative integers
- Decimal values
- Non-numeric values

must not be accepted as valid course capacity.

FR-004: Blank Capacity

A blank Maximum Students value submitted through the Admin Course form must be stored as `NULL`.

FR-005: Existing Courses

Existing courses must continue to work with unlimited capacity after the database migration.

6.2 Course API Response

FR-006: Maximum Students Response

Course API responses must include:

```
max_students
```

FR-007: Enrolled Count Response

Course API responses must include:

```
enrolled_count
```

The count must represent the current approved enrollment count used for capacity enforcement.

FR-008: Seats Remaining Response

Course API responses must include:

```
seats_remaining
```

For a limited-capacity course:

```
seats_remaining = max_students - enrolled_count
```

For an unlimited-capacity course, the response must represent unlimited availability appropriately.

FR-009: Full Status Response

Course API responses must include:

```
is_full
```

A limited-capacity course is full when:

```
enrolled_count >= max_students
```

An unlimited-capacity course must not be considered full.

FR-010: Consistent Course Availability

The same capacity rules must be used consistently by the Home page, Course Catalogue, Course Details, Admin Course Management, and enrollment backend.

6.3 Admin Course Management

FR-011: Maximum Students Field

The Admin Course form must include a Maximum Students field.

FR-012: Optional Maximum Students

The Maximum Students field must be optional.

FR-013: Capacity Validation

The backend must validate that a provided capacity is a positive integer.

Frontend validation may be added for usability, but backend validation must remain authoritative.

FR-014: Reduce Capacity Validation

An administrator must not be allowed to reduce course capacity below the current approved enrollment count.

FR-015: Equal Capacity Allowed

An administrator must be allowed to set capacity equal to the current approved enrollment count.

For example:

```
Approved enrollments: 10  
New capacity: 10
```

must be allowed.

FR-016: Unlimited Capacity

An administrator must be able to clear the Maximum Students field to return the course to unlimited capacity.

FR-017: Invalid Capacity Response

When an administrator submits an invalid capacity, the backend must reject the request with an appropriate validation error.

6.4 Course Availability Display

FR-018: Home Page Availability

The Home page must display course availability.

FR-019: Course Catalogue Availability

The Course Catalogue must display course availability.

FR-020: Course Details Availability

Course Details must display course availability.

FR-021: Admin Course Management Availability

Admin Course Management must display course availability.

FR-022: Capacity Display Format

For a limited-capacity course, availability must be displayed in the format:

```
15 / 30 students
```

where:

- 15 is the current approved enrollment count.
- 30 is the maximum capacity.

FR-023: Unlimited Display

When no capacity is configured, the interface must display:

FR-024: Course Full Display

When the course has no remaining seats, the frontend must display:

Course Full

FR-025: Full Course and Existing Enrollment

The frontend must not confuse a full course with a course in which the current student is already enrolled.

The UI must correctly distinguish:

- Available course.
- Full course.
- Already-enrolled course.
- Unlimited course.

6.5 Enrollment Capacity Enforcement**FR-026: Capacity Check**

Before approving a new enrollment, the backend must check the course capacity.

FR-027: Unlimited Enrollment

When `max_students` is `NULL`, the capacity restriction must not prevent enrollment.

FR-028: Full Course Enrollment

When the approved enrollment count has reached the maximum capacity, a new enrollment must not be approved.

FR-029: Full Course HTTP Status

When enrollment is attempted for a full course, the backend must return:

HTTP 409 Conflict

FR-030: Full Course Message

The HTTP 409 response must provide a clear message indicating that the course is full.

FR-031: Atomic Enrollment

The capacity check and enrollment insertion must occur atomically.

FR-032: Concurrent Enrollment Protection

Two students attempting to obtain the final available seat concurrently must not both succeed.

FR-033: Capacity Cannot Be Exceeded

Under concurrent enrollment requests, the number of approved enrollments must never exceed `max_students`.

FR-034: Server-Side Enforcement

Capacity enforcement must be performed by the backend/database transaction and must not depend solely on frontend availability checks.

6.6 Enrollment Deletion

FR-035: Seat Released After Deletion

When an enrollment is deleted, the corresponding seat must become available again.

FR-036: Recalculated Availability

After an enrollment is deleted, the course's `enrolled_count`, `seats_remaining`, and `is_full` values must reflect the updated enrollment state.

FR-037: Re-Enrollment After Seat Release

When a seat becomes available after enrollment deletion, another student must be able to enroll provided all other enrollment requirements are satisfied.

6.7 Seed Data

FR-038: Limited-Capacity Seed Course

The seed data must contain at least one course with a small capacity suitable for testing.

FR-039: Unlimited Seed Compatibility

Existing seed courses that do not specify capacity must remain unlimited.

7. Security and Data Integrity Requirements

SEC-001: Server-Side Capacity Enforcement

Course capacity must be enforced on the backend.

SEC-002: No Frontend-Only Protection

Frontend availability checks must not be treated as the authoritative enrollment restriction.

SEC-003: Atomic Capacity Enforcement

The capacity check and enrollment insertion must be protected by a database transaction or equivalent atomic mechanism.

SEC-004: Concurrent Request Protection

Concurrent enrollment requests must be serialized or otherwise protected so that capacity cannot be exceeded.

SEC-005: Capacity Validation

The backend must reject invalid maximum student values.

SEC-006: Positive Capacity

A configured capacity must be greater than zero.

SEC-007: Administrator Capacity Protection

An administrator must not be able to set a capacity below the current approved enrollment count.

SEC-008: Approved Enrollment Count

Capacity enforcement must use the appropriate approved enrollment count rather than an arbitrary or client-supplied value.

SEC-009: Client Value Trust

The backend must not trust `enrolled_count`, `seats_remaining`, or `is_full` values supplied by the client.

SEC-010: Database Integrity

The implementation must ensure that concurrent requests cannot produce a state where approved enrollments exceed the configured capacity.

8. Configuration / Potentially Affected Components

Database

Potentially affected areas include:

- `backend/schema.sql`
- New database migration file
- `backend/seed.sql`

The database changes include:

- Adding `max_students INT NULL`.
- Supporting unlimited capacity through `NULL`.
- Maintaining compatibility with existing courses.
- Providing test seed data with a limited-capacity course.

Backend

Potentially affected areas include:

- `backend/models/courseModel.js`
- `backend/models/enrollmentModel.js`
- `backend/controllers/courseController.js`
- `backend/controllers/enrollmentController.js`

The backend changes include:

- Reading and writing `max_students`.
- Validating maximum student capacity.
- Calculating approved enrollment count.
- Calculating remaining seats.
- Calculating `is_full`.
- Returning availability information through course responses.
- Preventing enrollment when capacity is reached.
- Returning HTTP 409 for full courses.
- Implementing transactional enrollment logic.
- Protecting concurrent enrollment attempts.
- Validating administrator capacity reductions.
- Releasing seats when enrollments are deleted.

Frontend

Potentially affected areas include:

- `frontend/src/pages/ManageCourses.jsx`
- `frontend/src/pages/CourseDetails.jsx`
- `frontend/src/pages/Courses.jsx`
- `frontend/src/components/CourseCard.jsx`
- `frontend/src/pages/Home.jsx`
- `frontend/src/App.css`

The frontend changes include:

- Maximum Students input in Admin Course Management.
- Course availability display.
- Capacity display in the required format.
- Unlimited display.
- Course Full display.
- Correct handling of already-enrolled courses.
- Full-course enrollment messaging.
- Styling for availability and full-course states.

Expected Files

The expected files are:

- `backend/schema.sql`
- New database migration file
- `backend/seed.sql`
- `backend/models/courseModel.js`
- `backend/models/enrollmentModel.js`
- `backend/controllers/courseController.js`
- `backend/controllers/enrollmentController.js`
- `frontend/src/pages/ManageCourses.jsx`
- `frontend/src/pages/CourseDetails.jsx`
- `frontend/src/pages/Courses.jsx`
- `frontend/src/components/CourseCard.jsx`
- `frontend/src/pages/Home.jsx`
- `frontend/src/App.css`

Developers must perform an impact analysis before implementation.

9. Constraints / Assumptions

- `max_students` must be nullable.
- `NULL` means unlimited capacity.
- A configured capacity must be a positive integer.
- Blank capacity means unlimited capacity.
- Existing courses must continue to operate as unlimited courses.
- Capacity must be enforced by the backend.
- Frontend availability checks must not be the only protection against over-enrollment.
- Approved enrollment count must be used for capacity enforcement.
- A course with capacity 2 may have at most two approved enrollments.
- A third enrollment attempt must be rejected.
- A full course must return HTTP 409 during enrollment.
- The HTTP 409 response must clearly indicate that the course is full.
- The capacity check and enrollment insertion must be atomic.
- Concurrent enrollment attempts must not exceed capacity.
- Deleting an enrollment must release the associated seat.
- Administrators must not be able to reduce capacity below the current approved enrollment count.
- Administrators may set capacity equal to the current approved enrollment count.
- Administrators may clear capacity to restore unlimited capacity.
- At least one seeded course must have a small capacity for testing.
- Course responses must expose `max_students`, `enrolled_count`, `seats_remaining`, and `is_full`.
- The frontend must display availability on Home, Course Catalogue, Course Details, and Admin Course Management.
- Limited capacity must be displayed using the `current / maximum students` format.
- Unlimited courses must display `Unlimited`.
- Full courses must display `Course Full`.
- A full course must not be confused with a course where the current student is already enrolled.
- Existing enrollment functionality must continue to work.
- Existing course functionality must continue to work.
- Database-level or transactional protection must be authoritative for concurrent enrollment.

10. Acceptance Criteria

AC-001: Unlimited Courses Remain Unlimited

Courses with `max_students = NULL` continue to allow enrollment without a capacity restriction.

AC-002: Maximum Students Column

The database contains the `max_students INT NULL` column.

AC-003: Positive Capacity

A course can only have a positive integer as its configured capacity.

AC-004: Blank Capacity

Submitting a blank Maximum Students field stores the capacity as `NULL`.

AC-005: Existing Courses

Existing courses continue to work as unlimited-capacity courses after the migration.

AC-006: Capacity 2 Allows Two Students

A course with capacity 2 allows only two approved enrollments.

AC-007: Third Enrollment Rejected

A third student attempting to enroll in a course with two approved enrollments is rejected.

AC-008: Full Course Returns 409

An enrollment attempt for a full course returns HTTP 409 Conflict.

AC-009: Clear Full-Course Message

The full-course response provides a clear message indicating that the course is full.

AC-010: Course Response Contains Capacity

Course responses include `max_students`.

AC-011: Course Response Contains Enrollment Count

Course responses include `enrolled_count`.

AC-012: Course Response Contains Remaining Seats

Course responses include `seats_remaining`.

AC-013: Course Response Contains Full Status

Course responses include `is_full`.

AC-014: Correct Remaining Seats

For a course with capacity 30 and 15 approved enrollments, `seats_remaining` is 15.

AC-015: Full Status Calculation

A course is marked `is_full = true` when approved enrollments reach the configured capacity.

AC-016: Unlimited Full Status

An unlimited course is not marked as full.

AC-017: Home Page Availability

The Home page displays the correct course availability.

AC-018: Catalogue Availability

The Course Catalogue displays the correct course availability.

AC-019: Course Details Availability

Course Details displays the correct course availability.

AC-020: Admin Availability

Admin Course Management displays the correct course availability.

AC-021: Capacity Display Format

A limited course displays availability in the format:

15 / 30 students

AC-022: Unlimited Display

A course without a configured capacity displays:

Unlimited

AC-023: Full Course Display

A course with no remaining seats displays:

Course Full

AC-024: Full Course Not Confused With Enrollment

The UI correctly distinguishes between a full course and a course in which the current student is already enrolled.

AC-025: Maximum Students Field

Admin Course Management provides a Maximum Students field.

AC-026: Invalid Capacity Rejected

Zero, negative, decimal, or otherwise invalid capacity values are rejected.

AC-027: Capacity Below Enrollment Rejected

An administrator cannot reduce capacity below the current approved enrollment count.

AC-028: Capacity Equal to Enrollment Allowed

An administrator can set capacity equal to the current approved enrollment count.

AC-029: Capacity Can Be Cleared

An administrator can clear the capacity and return the course to unlimited capacity.

AC-030: Atomic Enrollment

The capacity check and enrollment insertion are performed atomically.

AC-031: Concurrent Enrollment Protection

Two students attempting to obtain the final seat concurrently cannot both successfully enroll.

AC-032: Capacity Never Exceeded

Concurrent enrollment requests cannot cause approved enrollments to exceed the configured capacity.

AC-033: Enrollment Deletion Frees Seat

Deleting an enrollment increases the available capacity by one where applicable.

AC-034: Availability Recalculated After Deletion

After an enrollment is deleted, `enrolled_count`, `seats_remaining`, and `is_full` reflect the updated state.

AC-035: Re-Enrollment After Seat Release

After a seat is released, another student can successfully enroll if the course is otherwise available.

AC-036: Limited-Capacity Seed Data

At least one seeded course has a small capacity suitable for testing.

AC-037: Backend Enforcement

Capacity restrictions remain effective even if a client attempts to bypass frontend availability checks.

AC-038: Approved Enrollment Count

Capacity enforcement is based on the current approved enrollment count.

AC-039: No Over-Enrollment

The system never allows approved enrollments to exceed the configured course capacity.

AC-040: Existing Enrollment Functionality

Existing enrollment functionality continues to work correctly for both unlimited and limited-capacity courses.

AC-041: Existing Course Functionality

Existing course functionality continues to work after the database and API changes.

AC-042: Migration Compatibility

The database migration can be applied without breaking existing course records.

AC-043: Full Course Cannot Be Enrolled Through Race Condition

A race condition cannot result in more approved enrollments than the configured capacity.

AC-044: Course Availability Is Consistent

The availability displayed across Home, Course Catalogue, Course Details, and Admin Course Management matches the backend course state.

11. Expected Outcome

After implementation:

- Courses support an optional maximum student capacity.
- `NULL` capacity represents unlimited enrollment.
- Configured capacity must be a positive integer.
- Existing courses remain unlimited unless a capacity is explicitly configured.
- Administrators can configure course capacity through the Maximum Students field.
- Blank capacity restores unlimited enrollment.
- Course APIs return `max_students`, `enrolled_count`, `seats_remaining`, and `is_full`.
- Course availability is displayed on Home, Course Catalogue, Course Details, and Admin Course Management.
- Limited courses display availability such as 15 / 30 students.
- Unlimited courses display `Unlimited`.
- Full courses display `Course Full`.
- A course with capacity 2 permits only two approved enrollments.
- A third student cannot enroll in a full course.
- Full-course enrollment attempts return HTTP 409.
- Full-course messages are clear to the user.
- A full course is not confused with an already-enrolled course.
- Administrators cannot reduce capacity below the current approved enrollment count.
- Administrators can set capacity equal to the current enrollment count.
- Capacity checks and enrollment insertion are atomic.
- Concurrent enrollment attempts cannot exceed the configured capacity.
- Deleting an enrollment frees the corresponding seat.

- Course availability is recalculated after enrollment deletion.
- At least one seeded course has a small capacity for testing.
- Backend/database enforcement prevents frontend bypasses and race-condition over-enrollment.
- Existing enrollment and course functionality continues to work correctly.